

DESH VROMON

Software Requirements Specification

*Intelligent Tourism Management Platform with Seasonal Analytics,
Accommodation, Transportation, and Personalized Travel Planning*

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of **Desh Vromon**, an intelligent tourism management platform designed to simplify tourism discovery, travel planning, service selection, and trip management.

This document establishes the system scope, objectives, user roles, functional requirements, non-functional requirements, constraints, assumptions, and proposed system architecture. It will serve as the primary reference for the design, implementation, and testing phases of the project.

1.2 Project Overview

Desh Vromon is a web-based tourism management platform initially focused on tourism within Bangladesh. The platform integrates destination discovery, seasonal tourism analytics, accommodation, transportation, tourism activities, personalized travel planning, itinerary generation, budget estimation, and service management into a unified system.

The platform is designed to serve three major categories of users: travelers, tourism service providers, and administrators. Each user category will have appropriate functionality based on its role and responsibilities.

2. Problem Statement

Tourism-related information is frequently distributed across different websites, social media platforms, travel agencies, hotel pages, transportation services, and informal sources. Travelers often need to consult multiple sources to determine where to travel, when to travel, where to stay, how to travel, what activities to perform, and how much a trip may cost.

This fragmented process makes travel planning time-consuming, inconsistent, and difficult to manage.

Desh Vromon addresses this problem by providing a centralized platform that integrates tourism information, seasonal insights, accommodation, transportation, activities, budgeting, itinerary planning, and tourism services into a single system.

3. Project Scope

3.1 In-Scope Features

The initial version of Desh Vromon will include the following functionalities:

- User registration, authentication, and profile management.
- Destination discovery and detailed destination information.
- Seasonal tourism analytics and recommended travel periods.
- Accommodation search, information, and supported booking.
- Transportation information and supported booking facilities.
- Tourism activities and tour packages.
- Personalized trip planning and itinerary generation.
- Travel budget estimation and expense comparison.
- Reviews and ratings for eligible destinations and services.
- Tourism service provider management.

- Administrative management and content moderation.
- Basic tourism analytics and reporting.

3.2 Out-of-Scope Features

The initial version will not include international tourism management, direct operation of hotels or transportation services, an independent payment gateway, real-time transportation control, or full-scale commercial deployment.

These capabilities may be considered as part of future versions of the platform.

4. Project Objectives

The major objectives of Desh Vromon are:

1. **Centralize Tourism Information:** Provide destination, accommodation, transportation, activity, and travel information through a single platform.
2. **Improve Travel Decision-Making:** Help users select suitable destinations based on season, budget, duration, interests, and personal preferences.
3. **Provide Seasonal Tourism Insights:** Present recommended travel periods, seasonal conditions, estimated costs, and available tourism activities.
4. **Simplify Travel Planning:** Integrate destinations, accommodation, transportation, activities, itineraries, and expenses into one planning process.
5. **Provide Personalized Recommendations:** Recommend suitable destinations, activities, accommodation, and transportation options according to user preferences.
6. **Support Tourism Service Providers:** Provide digital management facilities for hotels, resorts, transportation providers, guides, and tour operators.
7. **Provide Tourism Analytics:** Generate insights regarding destination popularity, seasonal demand, bookings, and tourism trends.
8. **Ensure Future Scalability:** Design the system so that additional destinations, services, integrations, and intelligent features can be incorporated later.

5. System Users and Roles

5.1 Traveler

Travelers are the primary users of the platform. They can search and explore destinations, view seasonal information, find accommodation and transportation, discover activities, create travel plans, generate itineraries, estimate budgets, make supported bookings, and submit reviews and ratings.

5.2 Tourism Service Provider

Service providers may include hotels, resorts, transportation providers, tour operators, local guides, and activity providers. They can register and manage their services, update pricing and availability, manage bookings, and maintain service-related information.

5.3 Administrator

Administrators are responsible for managing users, destinations, service providers, bookings, reviews, tourism content, and system analytics. Administrators may also verify service providers and moderate platform content.

6. System Architecture

Desh Vromon follows a layered web-based architecture designed to provide modularity, scalability, maintainability, and separation of responsibilities. The proposed architecture consists of three primary layers: the Presentation Layer, Application/Service Layer, and Data Layer. External services are integrated with the application through appropriate APIs.

6.1 Architecture Overview

The overall system architecture is illustrated in Figure 1.

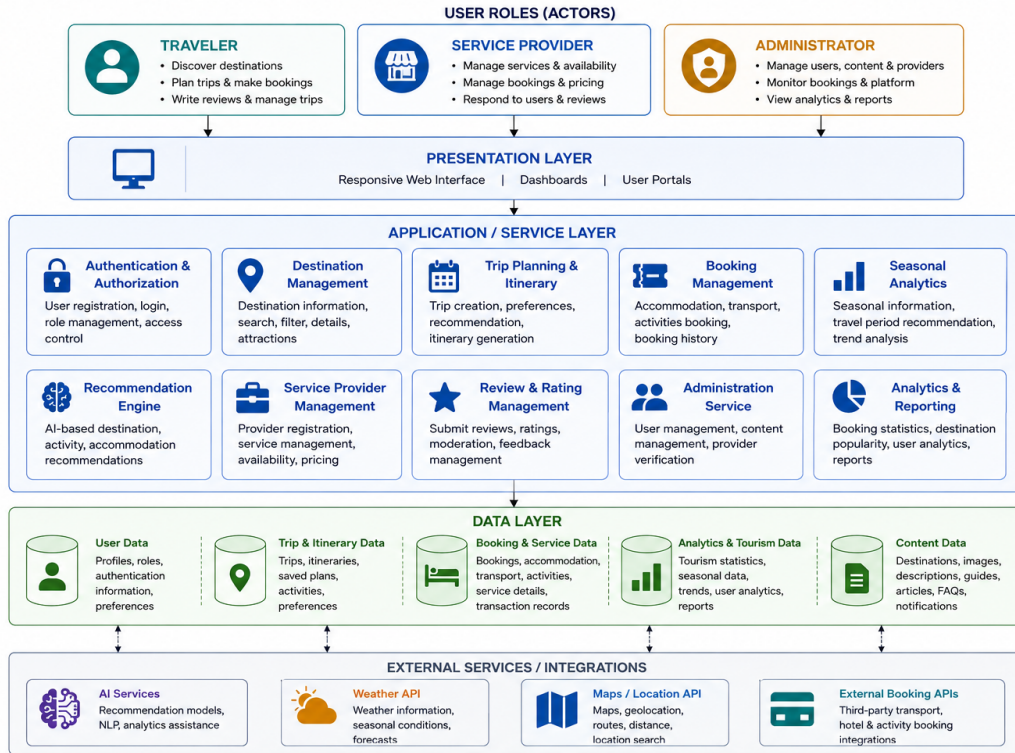


Figure 1: Proposed System Architecture of Desh Vromon

Figure 1: **Proposed System Architecture of Desh Vromon**

The architecture follows a structured request-processing flow. Travelers, service providers, and administrators interact with the system through the Presentation Layer. User requests are then processed by the Application/Service Layer, which contains the core business logic of the platform. The required information is retrieved from or stored in the Data Layer.

The Application/Service Layer may also communicate with external services when additional information or functionality is required.

6.2 Presentation Layer

The Presentation Layer provides the user-facing interface of the platform. It consists of the responsive web interface, user portal, and role-specific dashboards.

The layer provides different interfaces for:

- Travelers
- Tourism service providers

- Administrators

It is responsible primarily for presenting information and receiving user interactions.

6.3 Application/Service Layer

The Application/Service Layer contains the core business logic of Desh Vromon. Major services include:

- **Authentication and Authorization:** Handles user authentication, authorization, and role-based access.
- **Destination Management:** Manages destinations, attractions, tourism information, and destination-related content.
- **Trip Planning and Itinerary:** Handles travel plans, itineraries, activities, and personalized trip organization.
- **Booking Management:** Manages accommodation, transportation, and supported booking operations.
- **Seasonal Analytics:** Processes seasonal tourism information and generates travel-period insights.
- **Recommendation Engine:** Generates personalized recommendations based on user preferences, budget, duration, interests, and available tourism information.
- **Service Provider Management:** Allows tourism service providers to manage their services, availability, pricing, and relevant bookings.
- **Administration Service:** Provides system-level management, content moderation, user management, and administrative analytics.

6.4 Data Layer

The Data Layer is responsible for persistent storage and retrieval of system information. It may contain the following major categories of data:

- User and profile data
- Destination and tourism data
- Trip and itinerary data
- Accommodation and transportation data
- Booking and service data
- Reviews and ratings
- Seasonal and tourism analytics data

The Data Layer is accessed through the Application/Service Layer to maintain proper separation between business logic and data storage.

6.5 External Services

Desh Vromon may integrate with external services to extend the functionality of the platform. These services include:

- **AI Services:** Support personalized recommendations and intelligent travel planning.
- **Weather API:** Provides weather and seasonal information that can contribute to travel recommendations.
- **Maps and Location API:** Provides geographic, mapping, location, and route-related function-

ality.

- **External Booking APIs:** May provide accommodation or transportation availability and booking-related information.

External services are treated as independent components and are accessed through the Application/Service Layer rather than directly from the user interface.

7. Functional Requirements

The functional requirements define the specific services and operations that the Desh Vromon system shall provide to its users.

7.1 Authentication and User Management

ID	Requirement	Description
FR-01	User Registration	The system shall allow new users to create an account using the required personal information and credentials.
FR-02	User Authentication	The system shall allow registered users to securely log in and log out of the platform.
FR-03	Profile Management	Users shall be able to view and update their personal profile information.
FR-04	Role-Based Access Control	The system shall provide appropriate access permissions based on user roles such as Traveler, Service Provider, and Administrator.

Table 1: Functional Requirements for Authentication and User Management

7.2 Destination Management

ID	Requirement	Description
FR-05	Destination Discovery	Users shall be able to browse and explore available tourist destinations.
FR-06	Destination Search	Users shall be able to search for destinations using keywords and relevant criteria.
FR-07	Destination Filtering	Users shall be able to filter destinations based on location, budget, season, destination type, and available activities.
FR-08	Destination Details	The system shall provide detailed information about destinations, including attractions, activities, estimated duration, budget, and recommended travel periods.
FR-09	Destination Content Management	Administrators shall be able to add, update, and manage destination-related information.

Table 2: Functional Requirements for Destination Management

7.3 Seasonal Tourism Analytics

ID	Requirement	Description
FR-10	Seasonal Information	The system shall provide seasonal tourism information for available destinations.
FR-11	Recommended Travel Period	The system shall identify suitable travel periods for destinations based on available seasonal information.
FR-12	Season Comparison	Users shall be able to compare tourism conditions across different months or seasons where sufficient data is available.
FR-13	Tourism Demand Analysis	The system shall provide information regarding destination popularity and seasonal tourism demand where sufficient data is available.
FR-14	Weather Integration	The system may retrieve weather-related information from external services to improve seasonal travel recommendations.

Table 3: Functional Requirements for Seasonal Tourism Analytics

7.4 Accommodation Management

ID	Requirement	Description
FR-15	Accommodation Search	Users shall be able to search for hotels, resorts, and other available accommodation options.
FR-16	Accommodation Filtering	Users shall be able to filter accommodation based on price, rating, location, facilities, room type, and availability.
FR-17	Accommodation Details	The system shall display accommodation information including pricing, facilities, location, images, ratings, and availability.
FR-18	Accommodation Booking	Users shall be able to make and manage supported accommodation bookings.
FR-19	Provider Accommodation Management	Verified accommodation providers shall be able to add, update, and manage their accommodation information.

Table 4: Functional Requirements for Accommodation Management

7.5 Transportation Management

ID	Requirement	Description
FR-20	Transportation Search	Users shall be able to search for available transportation options for their trips.
FR-21	Transportation Information	The system shall provide route, schedule, estimated duration, transportation type, and cost information.
FR-22	Transportation Filtering	Users shall be able to filter transportation options according to type, price, duration, and other available criteria.
FR-23	Transportation Reservation	Users shall be able to reserve supported transportation services through the platform or integrated services.

Table 5: Functional Requirements for Transportation Management

7.6 Personalized Trip Planning

ID	Requirement	Description
FR-24	Create Travel Plan	Users shall be able to create and manage personalized travel plans.
FR-25	Travel Preferences	Users shall be able to specify destination, duration, number of travelers, budget, interests, and preferred travel style.
FR-26	Personalized Recommendations	The system shall recommend suitable destinations, activities, accommodation, and transportation based on user preferences.
FR-27	Itinerary Generation	The system shall generate a suggested itinerary based on destination, duration, preferences, budget, and available activities.
FR-28	Itinerary Modification	Users shall be able to modify, reorder, and save activities within their generated itineraries.

Table 6: Functional Requirements for Personalized Trip Planning

7.7 Budget and Expense Management

ID	Requirement	Description
FR-29	Budget Definition	Users shall be able to define a maximum budget for their planned trip.
FR-30	Expense Estimation	The system shall estimate transportation, accommodation, food, activity, and other travel-related expenses.
FR-31	Budget Comparison	The system shall compare estimated trip expenses with the user's defined budget.
FR-32	Expense Breakdown	The system shall provide a categorized breakdown of estimated travel expenses.

Table 7: Functional Requirements for Budget and Expense Management

7.8 Booking Management

ID	Requirement	Description
FR-33	Booking Creation	Users shall be able to create supported accommodation and transportation bookings.
FR-34	Booking Status	Users shall be able to view the current status of their bookings.
FR-35	Booking History	Users shall be able to view their previous and current booking history.
FR-36	Booking Management	Service providers and administrators shall be able to manage relevant booking records.

Table 8: Functional Requirements for Booking Management

7.9 Reviews and Ratings

ID	Requirement	Description
FR-37	Service Rating	Eligible users shall be able to provide ratings for destinations and tourism services.
FR-38	Review Submission	Eligible users shall be able to submit reviews for destinations and services.
FR-39	Review Display	The system shall display relevant reviews and ratings to support user decision-making.
FR-40	Review Moderation	Administrators shall be able to review and moderate inappropriate or misleading content.

Table 9: Functional Requirements for Reviews and Ratings

7.10 Tourism Service Provider Management

ID	Requirement	Description
FR-41	Provider Registration	Tourism service providers shall be able to register on the platform.
FR-42	Provider Verification	Administrators shall be able to verify and approve registered service providers.
FR-43	Service Management	Verified providers shall be able to add, update, and manage their tourism services.
FR-44	Availability Management	Providers shall be able to update service availability and relevant pricing information.
FR-45	Provider Booking Management	Providers shall be able to view and manage bookings related to their services.

Table 10: Functional Requirements for Tourism Service Provider Management

7.11 Administration and Analytics

ID	Requirement	Description
FR-46	User Management	Administrators shall be able to manage registered users and their access permissions.
FR-47	Content Management	Administrators shall be able to manage destinations, tourism content, and platform information.
FR-48	Service Provider Management	Administrators shall be able to manage and monitor registered service providers.
FR-49	Booking Monitoring	Administrators shall be able to monitor booking activities and relevant system records.
FR-50	Tourism Analytics Dashboard	The system shall provide administrators with analytical information regarding destination popularity, seasonal trends, and booking activities.

Table 11: Functional Requirements for Administration and Analytics

8. Non-Functional Requirements

- **Performance:** The system should provide responsive interaction and efficient retrieval of tourism information.
- **Security:** User credentials and sensitive information shall be protected through appropriate authentication, authorization, and secure data-handling mechanisms.
- **Usability:** The platform shall provide an intuitive, consistent, and user-friendly interface.
- **Scalability:** The architecture should support an increasing number of users, destinations, services, and bookings.
- **Reliability:** The system shall maintain data consistency and provide appropriate error handling.
- **Maintainability:** The modular architecture shall allow individual components to be modified and maintained with minimal impact on other components.
- **Compatibility:** The platform should support modern web browsers and responsive layouts for desktop, tablet, and mobile devices.

9. Constraints and Assumptions

9.1 Constraints

- The initial system will focus on tourism within Bangladesh.
- Tourism data availability may affect the accuracy of seasonal and analytical features.
- Real-time booking functionality may depend on third-party APIs or service providers.
- Advanced AI features may require external AI services or suitable datasets.
- Development is constrained by the available project timeline, team size, and technical resources.

9.2 Assumptions

- Users have internet access and a modern web browser.
- Tourism service providers provide accurate service information.
- Required tourism datasets are available for development and testing.
- External APIs remain accessible during system operation.
- Users provide accurate travel preferences and booking information.

10. AI-Assisted Requirement Engineering

AI tools will be used as supporting tools during the requirement engineering and documentation process. AI assistance may include:

- Initial requirement generation
- Requirement refinement and classification
- Identification of potentially missing requirements
- Use-case drafting
- Requirement consistency checking
- Initial task decomposition
- Documentation assistance

The requirement development workflow will follow:

Project Idea → Team Discussion → Initial Requirements → AI-Assisted Draft → Human Validation → Final SRS

AI-generated requirements will not be incorporated directly into the final specification without

review. Both team members will validate the requirements for correctness, feasibility, consistency, and relevance to the project scope.

11. Future Scope

Future versions of Desh Vromon may incorporate the following capabilities:

- International tourism support
- AI-based conversational travel assistant
- Real-time weather and travel alerts
- Predictive tourism demand analysis
- Dynamic pricing insights
- Real-time transportation integration
- Dedicated mobile applications
- Multilingual support
- Advanced tourism business analytics
- Integration with additional third-party booking platforms